Infineon/BIT Excellence Program 2026
Sustainable Supply Chain for Dried Mango from Africa to Germany

Sprint Note

Week of August 17–21, 2026 — Prepared by MADIEGA S Aida Justine

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Summary

Backend implementation for the Energy Supply Systems module started this week, following the tech stack confirmed by the team (Spring Boot, PostgreSQL, Lombok). The module's data model was translated into working code aligned with the repository's existing conventions, and an improvement to the app mock-up was identified for the What-If Scenario Analysis tab.

This Week's Progress

- Tech stack confirmed and applied: Spring Boot, PostgreSQL, Lombok, React (JavaScript, JSX)
- Repository conventions reviewed (architecture, module ownership, git workflow) to align this module's code with the rest of the team
- Backend scaffolding started for modules/energysupply/: entities and repositories in place, following the layered structure (controller / service / repository / entity / dto) used across all nine modules
- Mock-up improvement identified: Added dynamic charts (Recharts) to the What-If tab so scenario changes update the visualization in real time
- Currently building Spring Boot proficiency step by step; open to a technical sync if useful, as was arranged for other modules

Next Steps

- Continue backend implementation: service and controller layers for the Energy Component entity
- Verify local setup via Docker Compose once enough backend code is in place to test

Sprint Note

Week of August 24–28, 2026 — Prepared by MADIEGA S Aida Justine

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Summary

Finalization of the Energy Supply module backend: the entire scope requested by this week's feedback (*"Take the dummy data, save them to the database, connect to the dashboard"*) is now delivered. A complete package of Java files has been created, including all entities.

The Repository/Service/Controller layers, and an automated seeder that injects the Base scenario data into PostgreSQL at startup. The aggregated dashboard endpoint now returns values fetched from the database, and the foundation is ready to be merged into the main repository.

This Week's Progress

- Completed the remaining entities (Repository + Service + Controller) following the already validated model and the repository's conventions (layered architecture: controller / service / repository / entity / dto).
- Added a seeder (EnergySupplyDataSeeder) that automatically inserts at startup: 5 components, 6 daily consumption records, 1 monthly carbon metric, and 1 generator event directly addressing the need to inject the Excel data into the database.
- Developed controllers exposing full CRUD operations on the 6 tables, plus an aggregated endpoint dedicated to the dashboard:
  GET /api/energy/dashboard/summary → now returns real data (no more static mocks).
- Verified the consistency of Java files (no syntax or brace errors, everything is coherent).
- Identified a point of attention: two values (PV capacity in kWp and CO_2 avoided) were left as TODO in the seeder because the source Excel file was no longer accessible after the container reset.

Next Steps

- Prepare the package ready for deployment: merge the modules/energysupply/ folder, quick test via docker compose up -d --build, and validation of endpoint visibility in Swagger
- Confirm the two exact values from the Base scenario Excel file (search for // TODO: confirm from Excel in EnergySupplyDataSeeder.java) and update them before the demo.
- **Start** the frontend with react.